Reproducibility Checklist

Instructions for Authors:

This document outlines key aspects for assessing reproducibility. Please provide your input by editing this `.tex` file directly.

For each question (that applies), replace the "Type your response here" text with your answer.

Example: If a question appears as

```
\question{Proofs of all novel
claims are included}
{(yes/partial/no)}
Type your response here
```

you would change it to:

```
\question{Proofs of all novel
claims are included}
{(yes/partial/no)}
yes
```

Please make sure to:

- Replace ONLY the "Type your response here" text and nothing else.
- Use one of the options listed for that question (e.g., **yes**, **no**, **partial**, or **NA**).
- **Not** modify any other part of the `\question` command or any other lines in this document.

You can `\input` this .tex file right before `\end{document}` of your main file or compile it as a stand-alone document. Check the instructions on your conference's website to see if you will be asked to provide this checklist with your paper or separately.

1. General Paper Structure

1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA) yes

1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no) yes

1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no) yes

2. Theoretical Contributions

2.1. Does this paper make theoretical contributions? (yes/no) no

If yes, please address the following points:

2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no) NA

2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no) NA

2.4. Proofs of all novel claims are included (yes/partial/no) NA

2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no) NA

2.6. Appropriate citations to theoretical tools used are given (yes/partial/no) NA

2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA) NA

2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA) NA

3. Dataset Usage

3.1. Does this paper rely on one or more datasets? (yes/no) yes

If yes, please address the following points:

3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA) yes

3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA) NA

3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA) NA

3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA) yes

3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA) partial. 3DSSG/3RScan-derived assets are available through their dataset terms; large scans and source checkpoints are not redistributed by this paper.

3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisficing (yes/partial/no/NA) partial. The paper describes the official validation split, source prediction dependency, and closed-vocabulary evaluation mapping; licensed raw data must be obtained from the original providers.

4. Computational Experiments

4.1. Does this paper include computational experiments? (yes/no) yes

If yes, please address the following points:

4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA) partial. The paper and supplement identify the fixed operating points, score definitions, and selection criteria; no exhaustive hyperparameter search is claimed.

4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no) partial. The Code and Data Supplement includes RelCompat3D preprocessing, geometry-join, adapter, and metric code. Licensed scans, third-party source checkpoints, and large regenerable caches are excluded and documented as external dependencies.

4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no) partial. The anonymized Code and Data Supplement includes the new method, evaluation, bootstrap, audit, Docker/config, and compact-evidence paths. Third-party source repositories, licensed datasets, checkpoints, and large row-level outputs remain external dependencies.

4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no) partial. The submission includes an anonymized code package; the final public repository URL and release license will be fixed before publication.

4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no) partial. Core compatibility scoring, metrics, and protocol scripts carry implementation comments and artifact manifests; some source-specific adapters use compact runbook documentation rather than line-level paper references.

4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA) yes

4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no) yes. Experiments used an Intel Core Ultra 7 265KF CPU, 62 GiB RAM, an NVIDIA GeForce RTX 5090 with 32,607 MiB memory and driver 580.159.03, Linux 6.17, and Docker 29.6.1. Dockerfiles pin the main Python, NumPy, PyTorch/CUDA, PyG, and source-runtime versions.

4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no) yes

4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no) yes

4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no) yes

4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no) yes

4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA) partial. Final RelCompat3D scores, K grid, thresholds, bootstrap seed/count, denominators, and checkpoint hashes are included. Third-party source defaults are referenced through their fixed Docker/config manifests.